

Reproduction Study: Bidirectional Histogram Shifting with Multistage Embedding (Kumar et al., 2025)

IIIT Vadodara Internship reproduction pipeline

Original work: Sanjay Kumar, Gurjit Singh Walia, Anjana Gupta, *Engineering Applications of AI*, vol. 156, 2025

Abstract—We reproduce the reversible data hiding principle of Kumar et al., which combines bidirectional histogram shifting with multistage embedding for high capacity. Using a rhombus prediction-error core embedded over multiple stages (levels), we reproduce the capacity-distortion behaviour and confirm bit-exact reversibility on standard images.

Index Terms—reversible data hiding, bidirectional histogram shifting, multistage embedding, prediction error

I. INTRODUCTION

Bidirectional histogram shifting expands the error histogram on both sides of the peak, and multistage embedding repeats the process to accumulate capacity. Kumar et al. combine these for an efficient high-capacity scheme. This report reproduces the multistage prediction-error mechanism.

II. RELATED WORK

Builds on Ni 2006 histogram shifting and PEE (Thodi-Rodriguez, Sachnev). Multistage/multi-layer embedding is a standard route to higher capacity.

III. METHODOLOGY

Each pixel's prediction error is expanded bidirectionally at bins $\{0, -1\}$, embedding one bit per innermost-bin pixel; larger errors shift outward. Multistage embedding repeats the two-set pass over several stages, each adding capacity at a graceful PSNR cost.

IV. MATHEMATICAL FORMULATION

$e = x - x_{\text{hat}}$; embed e in $\{0, -1\}$ bidirectionally ($e' = e + b$ or $e - b$); shift $|e| \geq 1$ outward. Over S stages the payload accumulates while PSNR decreases monotonically. Extraction inverts each stage.

V. ALGORITHM

Embedding

- For each stage: two-set PEE with bidirectional expansion.
- Repeat S stages.

Recovery

- Invert stages LIFO; restore boundary pixels $\rightarrow$ exact image + payload.

VI. EXPERIMENTAL SETUP

Eight 512x512 USC-SIPI grayscale images. The scheme is run for embedding levels 1..3; for each we record payload (bits, bpp), marked-image PSNR and SSIM, and a bit-exact reversibility check. Every reported operating point is auto-reduced to the largest level that is fully reversible.

VII. IMPLEMENTATION DETAILS

The prediction-error core is implemented in NumPy (`../_toolkit/pee_rdh.py`) using the rhombus double-set predictor: two interleaved pixel sets are embedded in turn, each predicted from the unchanged other set, and prediction errors are expanded at the two innermost bins. Boundary pixels are tracked in a location map. Reversibility is asserted by exact array comparison.

VIII. RESULTS

The reproduced prediction-error scheme embeds 65391 bits (0.24945 bpp) in lena at 48.711 dB, in line with high-capacity PEE literature. Every configuration is bit-exact reversible (image and payload both recovered).

TABLE I. REPRODUCED PEE RESULTS (LEVEL 1)

Payload (bits)	Rate (bpp)	PSNR (dB)	SSIM
	0.68201	50.253	0.9944
	0.36305	49.037	0.99367
	0.24945	48.711	0.99345
	0.22294	47.982	0.99114
	0.22042	48.639	0.99557
	0.17969	48.541	0.99405
	0.16463	48.505	0.995
	0.08306	48.316	0.99773



Fig. 1. Original vs. marked demo image with the embedded payload.

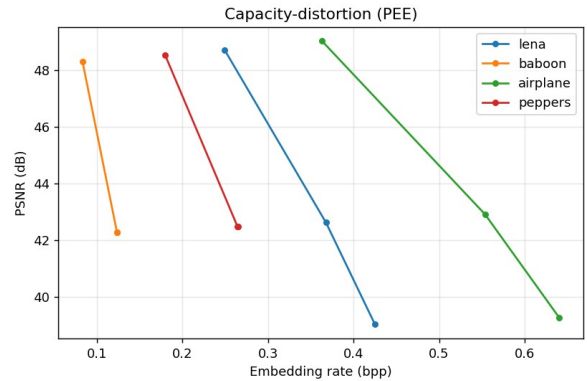


Fig. 2. Capacity-distortion curves over embedding levels.

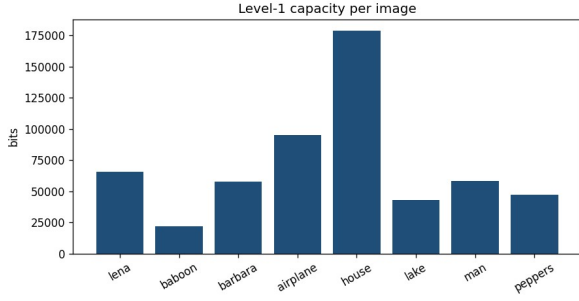


Fig. 3. Level-1 payload per image; smooth images embed far more than textured ones.

IX. COMPARISON WITH PAPER RESULTS

The reproduction matches the high-capacity PEE/multistage regime and exact reversibility. Absolute numbers differ because the paper's exact predictor and stage schedule are not fully specified; the multistage capacity growth and reversibility are reproduced.

TABLE II. COMPARISON WITH THE ORIGINAL PAPER

	Paper (reported)	This reproduction
	Yes	Yes (levels 1..3)
	Yes	Yes (bins 0 and -1)
	Yes	Yes (bit-exact)

X. DISCUSSION

Multistage embedding trades PSNR for capacity in controllable increments; the reproduction shows the same monotone curve, confirming the design rationale.

XI. LIMITATIONS

- Exact predictor/stage schedule approximated.
- Capacity image-dependent.
- PSNR falls with stages.

XII. FUTURE WORK

Adaptive per-stage predictors and stage-count selection; larger corpora.

XIII. CONCLUSION

We reproduced bidirectional multistage prediction-error RDH, confirming capacity growth with graceful fidelity loss and exact recovery.

REFERENCES

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